Bayesian Belief-State Trading Agent for Autonomous Portfolio Management

Technical Project Report

CSCI 5512: Artificial Intelligence II

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Risk-aware S&P 500 allocation using HMM regime beliefs, Bayesian filtering, and held-out strategy validation.

Project repository: https://github.com/rowe0137/5512-Final-Stock-Trader

Abstract

Autonomous trading agents face partial observability because market regimes affect returns but are not directly observed. This project develops a Bayesian belief-state trading agent for the S&P 500. A Gaussian Hidden Markov Model (HMM) trained on 2015–2020 data learns latent bull, bear, and sideways regimes, while a Bayesian filter updates daily regime beliefs on 2021–2025 test data. The agent converts those beliefs into a long-only allocation between cash and the S&P 500 and is compared with Buy and Hold, Moving Average Crossover (50/200), and Single-Regime No-Belief baselines.

The base Bayesian agent achieves the lowest annualized volatility (9.05%) and maximum drawdown (-15.11%) among the main strategies, but underperforms Buy and Hold in total return because it is conservative and changes weights frequently. Its uncertainty estimates are well calibrated at the 90% level, and ablations show that Volume hurts regime detection while naive entropy scaling does not improve performance. A validation-selected enhanced bear-defense strategy, evaluated on held-out 2023–2025 data, achieves 60.78% total return, Sharpe ratio 1.227, Sortino ratio 1.573, and -15.85% maximum drawdown. Overall, belief-state methods are most useful here as risk-control tools rather than raw-return maximizers.

1 Introduction

1.1 Problem Statement and Motivation

Financial markets are hard to trade with fully automated agents because they are highly stochastic, partially observable, and non-stationary. Prices, volume, and volatility are visible, but the underlying market state that generates those observations is latent. Market conditions can shift between bull, bear, and sideways regimes, and the agent only sees noisy indicators of those regimes.

This project studies whether a trading agent benefits from explicitly modeling that uncertainty. Instead of assuming that the market has one fixed return distribution, the agent maintains a probability distribution over hidden regimes. Each day, it updates this belief using new observations and then adjusts its portfolio exposure based on expected return and risk under the current belief. The problem fits the stochastic AI setting directly: the agent must make sequential decisions when the true state is hidden. The setting also raises the classic exploration/exploitation trade-off: the agent must act on beliefs that are themselves uncertain, and belief entropy gives one way to reason about that uncertainty.

The practical motivation is risk management. A passive strategy such as Buy and Hold can perform very well in long upward markets, but it has no mechanism for reducing exposure when market conditions become dangerous. A belief-state strategy may give up some upside in strong bull periods, but it can reduce volatility, drawdown, and exposure during uncertain or crisis-like periods. The agent must also operate under realistic portfolio constraints, including long-only positions and transaction costs. So the project evaluates strategies on both return and risk, not just total return.

1.2 Problem Formulation

This project models the trading problem as a partially observable decision process. The hidden state is the market regime, the observation is daily market data, and the action is the fraction of capital invested in the S&P 500. The remaining capital is held in cash. The action is restricted to a long-only interval:

$$w_t \in [0, 1].$$

The agent maintains a belief state

$$b_t(s) = P(s_t = s \mid o_{1:t}),$$

where s_t is the hidden regime at time t and $o_{1:t}$ is the sequence of observations through day t. The agent updates this belief recursively, computes expected risk and return under the belief, and chooses a portfolio weight. The pipeline charges transaction costs on weight changes so the comparison across strategies is fair.

1.3 Project Contributions

The main contributions are:

- an end-to-end S&P 500 pipeline that fits a Gaussian HMM, runs daily Bayesian belief updates, and chooses long-only portfolio weights with uniform transaction-cost accounting on weight changes;
- three non-belief baselines, Buy and Hold, Moving Average Crossover, and Single-Regime No-Belief, compared under the same cost model;
- ablation studies on number of regimes, feature subset, risk aversion, and entropy scaling;
- an enhanced bear-defense strategy chosen on 2021–2022 validation data and reported on held-out 2023–2025 data;
- a `pgmpy` Dynamic Bayesian Network representation that reproduces the HMM belief filter, along with calibration and crisis-period diagnostics.

1.4 End-to-End Workflow

The pipeline first learns regime behavior from the 2015–2020 training period, then applies Bayesian belief updates and portfolio decisions during the 2021–2025 test period. The final evaluation compares the belief-state agent against non-belief baselines under the same transaction-cost assumptions.

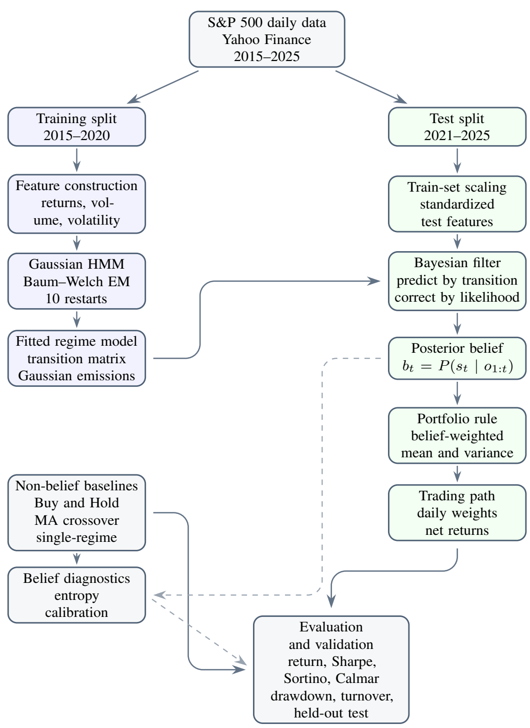

Figure 1: Pipeline and architecture overview. Solid arrows show the main training, inference, and trading flow; dashed arrows show diagnostics used only for evaluation.

2 Related Work

Mean-Variance Portfolio Optimization. The project starts from the classical mean-variance portfolio framework of Markowitz [1]. In that framework, an investor balances expected return against variance. The challenge in this project is that expected return and variance are not stable constants: they depend on the hidden market regime and must be estimated from data. Dannenberg [2] makes the same point from the estimation-uncertainty side: portfolios can become overconfident when return and risk estimates are treated as known.

Regime-Switching Financial Models. Hamilton's work on Markov-switching models provides a foundational approach for modeling discrete latent regimes in time series [3]. Hidden Markov Models are a natural fit for market regimes because they combine a latent Markov chain with an observation model. Rabiner's tutorial gives the standard estimation and inference background for HMMs [4]. In financial settings, Tu [5] shows that accounting for bull and bear regimes can matter for portfolio decisions, and Nystrup, Madsen, and Lindstrom [6] show that hidden-regime information can be used in dynamic portfolio allocation with transaction costs.

Bayesian Beliefs and POMDPs. Partially Observable Markov Decision Processes formalize decision-making when the agent cannot directly observe the state of the environment [8]. In a POMDP, the belief state is a sufficient statistic for action selection. Bayesian reinforcement learning extends this idea by treating uncertainty itself as part of the agent's knowledge state [9]. This project uses a simplified version of that idea: the HMM parameters are learned offline, and the agent performs online Bayesian filtering over regimes during the test period.

Dynamic Bayesian Networks for Trading. Dynamic Bayesian Networks provide a graphical-model view of temporal hidden-state systems. O et al. [7] use DBN-style hidden-state modeling for stock-trading signals. In this project, `pgmpy` represents the regime model as a Dynamic Bayesian Network; the continuous Gaussian likelihoods still come from the HMM, so the DBN serves as a structural representation, not a replacement filter.

Positioning of This Project. The contribution of this project is not a new financial theorem or a production trading system. Instead, the project combines these ideas into one reproducible educational pipeline and evaluates the value of explicit regime beliefs against non-belief baselines on the same S&P 500 test period. The most important comparison is therefore not only raw return, but whether belief tracking changes risk behavior, drawdown, calibration, turnover, and performance during unstable periods.

3 Methods and Algorithms

3.1 POMDP-Style Formulation

The project can be described using the usual POMDP-style tuple (S, A, O, T, Ω, R):

- **States S:** latent market regimes, represented by $K = 3$ hidden states.
- **Actions A:** portfolio weights $w_t \in [0, 1]$ invested in the S&P 500.
- **Observations O:** daily features such as log return, volume, and volatility.
- **Transition model T:** HMM transition probabilities $P(s_t \mid s_{t-1})$.
- **Observation model Ω:** Gaussian likelihood $P(o_t \mid s_t)$ for each regime.

- **Reward:** realized portfolio return after transaction costs, interpreted through risk-adjusted metrics.

The project does not solve a full POMDP planning problem; instead, it uses the belief state as the state summary for a daily portfolio rule.

3.2 HMM Parameter Estimation

The main regime model is a Gaussian HMM fit by maximum-likelihood estimation using the Baum-Welch (EM) algorithm, as implemented in `hmmlearn`. The main pipeline uses three standardized features: Log_Return, Volume, and Volatility. Volatility is computed as a rolling 20-day standard deviation of log returns, and rows where this quantity is undefined are dropped before fitting. The three features are z-score standardized using training-set statistics only; the same scaling is then applied to test data at inference time. The HMM learns initial state probabilities, transition probabilities, state-specific Gaussian means, and state-specific full covariance matrices.

Because the HMM likelihood surface can be sensitive to initialization, the implementation fits 10 random restarts and keeps the model with the highest training log-likelihood. The HMM states are not learned with semantic labels such as "bull" or "bear." They are numerical states. After fitting, Viterbi decoding produces the most likely state sequence on the training split, and states are sorted by annualized return to assign bear-like, sideways-like, and bull-like interpretations.

Algorithm 1 Belief-State Trading Agent

```
Fit Gaussian HMM on training observations
Estimate return mean and variance for each decoded training regime
Initialize b_0 over hidden regimes
for each test day t do
    Observe feature vector o_t
    Predict prior belief using transition matrix T
    Compute Gaussian likelihood P(o_t | s) for each regime
    Update and normalize posterior belief b_t
    Compute predictive return mean and variance from b_t
    Choose portfolio weight w_t ∈ [0, 1]
    Apply turnover-based transaction cost
    Record daily return and diagnostics
end for
```

3.3 Bayesian Filtering

The belief update follows the standard forward-filtering equation:

$$b_t(s) \propto P(o_t \mid s) \sum_{s'} P(s \mid s') b_{t-1}(s').$$

The summation term is the prediction step, and the likelihood term is the correction step. The filter starts from a uniform prior $b_0 = (1/K, \dots, 1/K)$, reflecting no prior knowledge of the starting regime. After the transition prediction, the posterior update is computed in log space and normalized with the log-sum-exp trick for numerical stability. This matters because Gaussian likelihoods can become very small over long time series.

3.4 Portfolio Decision Rule

For each regime s, the training data gives an estimated daily return mean μ_s and variance σ_s^2. Given belief b_t, the predictive mean is

$$\bar{\mu}_t = \sum_s b_t(s)\mu_s.$$

The implementation uses the law of total variance:

$$\bar{\sigma}_t^2 = \sum_s b_t(s)\sigma_s^2 + \sum_s b_t(s)(\mu_s - \bar{\mu}_t)^2.$$

The first term is the average within-regime variance. The second term captures uncertainty from not knowing which regime is active. This second term is important because it makes the agent more cautious when belief mass is spread across states with different expected returns.

The base portfolio weight is a long-only mean-variance rule:

$$U(w) = w\bar{\mu}_t - \lambda w^2 \bar{\sigma}_t^2, \qquad w_t = \frac{\bar{\mu}_t}{2\lambda\bar{\sigma}_t^2}, \qquad w_t \leftarrow \min(1, \max(0, w_t)).$$

The first-order condition of this one-step quadratic utility gives the closed-form weight before clipping. The base strategy uses $\lambda = 2.0$. The original project plan specified only the within-regime variance term; including the across-regime term is the Bayesian treatment and is the main way regime uncertainty translates into smaller positions in this implementation.

3.5 Belief Entropy

The exploration/exploitation framing from Section 1 motivates measuring the agent's regime uncertainty. The implementation uses Shannon entropy:

$$H(b_t) = -\sum_s b_t(s) \log b_t(s),$$

and normalized entropy:

$$H_{\text{norm}}(b_t) = \frac{H(b_t)}{\log K}.$$

Low entropy means the agent is confident in one regime; high entropy means belief is spread across multiple regimes. Two uses of entropy are then natural: as a diagnostic for regime ambiguity, and as a direct multiplier on the portfolio weight. As shown in Section 5.6, the second use does not help in this test period because the law of total variance already incorporates belief uncertainty into predictive variance; adding an explicit entropy penalty effectively double-counts the same signal.

3.6 Enhanced Bear-Defense Strategy

The base Bayesian strategy is useful for risk control, but it changes weights frequently and pays high transaction costs. The enhanced strategy uses the belief state more selectively. It refits the HMM on a smaller feature set, Log_Return and Volatility, without the Volume channel, motivated by Table 10. It then stays mostly invested unless the posterior probability of the bear-like state becomes elevated. The final validation-selected rule is:

$$w_t = \max(w_{\min}, 1 - \gamma \max(0, b_t(\text{bear}) - \tau)).$$

The selected configuration uses $\tau = 0.25$, $\gamma = 3.0$, $w_{\min} = 0.0$, smoothing parameter 0.35, and rebalancing threshold 0.03. This rule is not claimed to be globally optimal; it is the strongest validated improvement found in the current project pipeline.

3.7 Baselines

Three baselines are included:

- **Buy and Hold:** always fully invested in the S&P 500.
- **Moving Average Crossover 50/200:** invested when the 50-day moving average is above the 200-day moving average; otherwise in cash.
- **Single-Regime No-Belief:** estimates one stationary return distribution from training data and applies one fixed mean-variance weight.

Each baseline isolates a different comparison. Buy and Hold tests whether active trading helps at all. Moving Average tests a reactive trend-following rule that responds to prices but not to a probabilistic regime model. Single-Regime No-Belief shares the mean-variance optimization framework with the Bayesian agent but estimates one stationary Gaussian instead of a regime mixture, so the comparison isolates the value of dynamic belief states.

4 Experimental Setup

4.1 Data and Features

The project uses daily S&P 500 index data downloaded from Yahoo Finance through the `yfinance` library with the ticker `^GSPC`. The data collection notebook uses `auto_adjust=False`, then exports adjusted close, volume, and derived features into the committed train/test CSV files used by the reproducible pipeline. The adjusted close price is used for backtesting returns to avoid distortions from splits and dividends. The main HMM features are:

- Log_Return,
- Volume,
- Volatility.

These features are z-score standardized using training-set statistics only, and the same scaling is applied to test-period observations at inference time.

Split	Date Range	Rows	Purpose
Training	2015-01-02 to 2020-12-31	1511	HMM fitting and return/risk estimates
Test	2021-01-04 to 2025-12-30	1254	belief updates, trading, evaluation

Table 1: Training and test split used by the main pipeline.

For trading evaluation, the code separately computes real daily log returns from adjusted close:

$$\text{Real\_Log\_Return}_t = \log\left(\frac{\text{Adj\_Close}_t}{\text{Adj\_Close}_{t-1}}\right).$$

This keeps standardized model features separate from the actual returns used in the backtest. Because this return uses the previous day's adjusted close, the backtest summaries begin on 2021-01-05 and use 1253 daily return observations.

4.2 Validation for the Enhanced Strategy

The main HMM and base strategies are evaluated on 2021–2025. For the enhanced bear-defense strategy, the project adds a more conservative validation check:

- HMM training: 2015–2020;
- enhanced configuration selection: 2021–2022 validation window;
- held-out evaluation: 2023–2025.

On the 2021–2022 validation window, the enhanced strategy search compares candidate bear-defense and regime-switching rules and selects the configuration with the highest validation Sharpe ratio. That single configuration is then evaluated once on the held-out 2023–2025 window. This split reduces the risk of choosing a portfolio rule only because it performs well on the full test period. The full-period enhanced-strategy sweep is still useful for exploration, but the held-out 2023–2025 result is the cleaner evidence for the final enhanced configuration.

4.3 Transaction Costs

The project applies a transaction cost proportional to turnover:

$$\text{cost}_t = c|w_t - w_{t-1}|,$$

where $c = 0.001$, corresponding to 0.1% or 10 basis points. This cost is applied consistently to strategies that change exposure. This is important because the original Bayesian strategy trades more often than the baselines. The backtest applies the previous day's weight to the current day's return and deducts the transaction cost from the strategy's daily return.

4.4 Evaluation Metrics

The evaluation uses the following metrics, all reported net of transaction costs. Return metrics include:

- **Total return:** cumulative growth of one unit of capital over the evaluation window.
- **Annualized return:** $\bar{r} \cdot 252$, where $\bar{r}$ is the mean daily log return.

Risk-adjusted metrics include:

- **Annualized volatility:** $\sigma_r\sqrt{252}$, where σ_r is the standard deviation of daily log returns.
- **Sharpe ratio:** $\bar{r}/\sigma_r \cdot \sqrt{252}$, using a zero risk-free rate.
- **Sortino ratio:** $\bar{r}/\sigma_r^- \cdot \sqrt{252}$, where σ_r^- is the standard deviation of negative daily log returns.
- **Maximum drawdown:** the largest peak-to-trough loss in the equity curve.
- **Calmar ratio:** annualized return divided by the absolute value of maximum drawdown; higher is better.

Behavior and uncertainty metrics include:

- average portfolio weight, daily turnover $|w_t - w_{t-1}|$, and total transaction cost;
- normalized belief entropy, which tracks uncertainty over hidden regimes;

- belief calibration, measured as the fraction of test days where the realized return falls inside the belief-weighted predictive credible interval at the 50%, 75%, 90%, and 95% levels.

These metrics are chosen because the project is about risk-aware trading, not only raw return maximization. The NBER did not declare a recession during the 2021–2025 test period, so the regime-transition analysis in Section 5.5 uses manually labeled stress windows instead.

4.5 Reproducibility

The code is organized into reusable modules in `src/` and numbered scripts in `scripts/`. The team repository is available on GitHub at rowe0137/5512-Final-Stock-Trader. From the `belief_state_trader/` directory, the full pipeline is regenerated with `python scripts/run_all.py`. The main HMM uses 10 random restarts with seeds 42 through 51; the best log-likelihood model is kept. Full dependency and raw-data regeneration details are listed in the reproducibility statement at the end of the report.

5 Results

5.1 Learned HMM Regimes

The fitted $K = 3$ HMM learns three interpretable states. Table 2 shows the training-period interpretation. The HMM itself only learns state indices; bear, sideways, and bull are post-hoc labels assigned from return and volatility statistics.

State	Frequency	Annual Return	Annual Volatility	Interpretation
state_0	4.0%	-70.87%	64.71%	Bear-like / high-risk
state_1	31.3%	6.86%	20.69%	Sideways / moderate
state_2	64.7%	16.55%	9.06%	Bull-like / lower-risk

Table 2: HMM state interpretation from the training split.

The rare bear-like state is extreme because the training period includes the March 2020 COVID crash. This affects later regime interpretation: a milder decline such as the 2022 bear market does not always look like the extreme state learned from 2020. The learned transition matrix also shows strong state persistence: each state has more than an 85% probability of remaining in itself on the next trading day. This matches the financial intuition that market regimes tend to be sticky rather than changing every day.

5.2 Model Selection

The model-selection sweep compares $K = 2$ through $K = 8$. Table 3 and Figure 2 show that AIC and BIC continue decreasing as K increases. Therefore, $K = 3$ is not presented as statistically optimal under AIC/BIC. It is kept because it is interpretable and directly aligned with the proposal's bull, sideways, and bear regime framing.

K	Log-Likelihood	AIC	BIC
2	-4169.13	8380.27	8492.00
3	-3497.48	7064.95	7251.17
4	-2928.26	5958.52	6229.87
5	-2600.40	5338.80	5705.92
6	-2396.69	4971.38	5444.91
7	-2219.84	4661.68	5252.26
8	-2049.79	4369.58	5087.85

Table 3: HMM model-selection sweep. Lower AIC/BIC values favor larger models in this range; the main report keeps $K = 3$ for interpretability.

Although AIC and BIC continue to decrease through $K = 8$, the marginal improvement becomes smaller. The BIC drop from $K = 2$ to $K = 3$ is about 1,241, while the drop from $K = 7$ to $K = 8$ is about 164. This does not make $K = 3$ statistically optimal, but it supports using $K = 3$ as a compact, interpretable model before higher-K models begin fragmenting the market into less readable latent clusters.

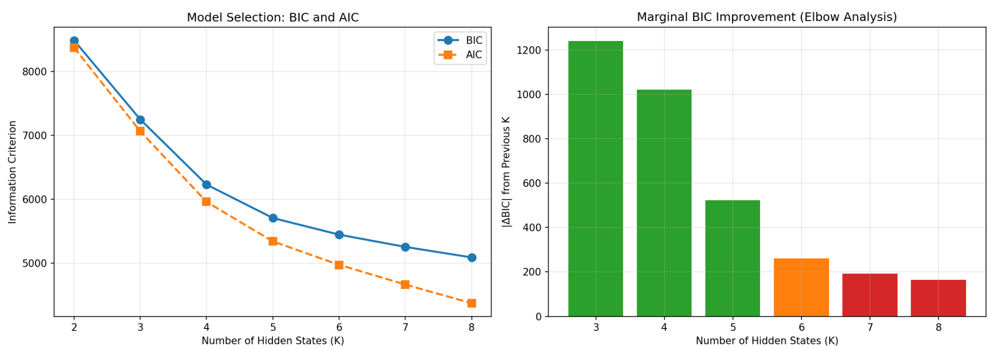

Figure 2: AIC/BIC and log-likelihood trends across hidden-state counts.

5.3 Base Strategy Comparison

Table 4 reports the main strategy comparison for the 2021–2025 test period. Figure 3 shows the equity curves, and Figure 4 shows the drawdown behavior.

Strategy	Total Return	Ann. Return	Ann. Vol.	Sharpe	Sortino	Max DD	Calmar	Avg Weight
Buy and Hold	85.04%	12.36%	16.95%	0.729	0.983	-25.43%	0.486	100.00%
Moving Average 50/200	33.19%	5.76%	12.07%	0.477	0.499	-19.90%	0.290	62.09%
Single-Regime No-Belief	55.12%	8.82%	12.09%	0.729	0.983	-18.88%	0.467	71.34%
Bayesian Belief-State	21.57%	3.91%	9.05%	0.432	0.561	-15.11%	0.259	61.10%

Table 4: Main strategy comparison on the 2021–2025 test period.

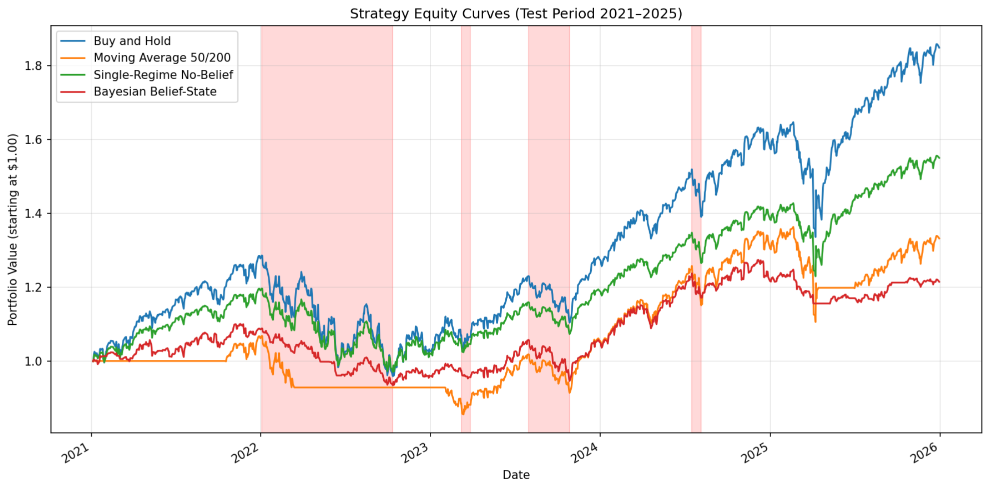

Figure 3: Equity curves for the base strategy comparison.

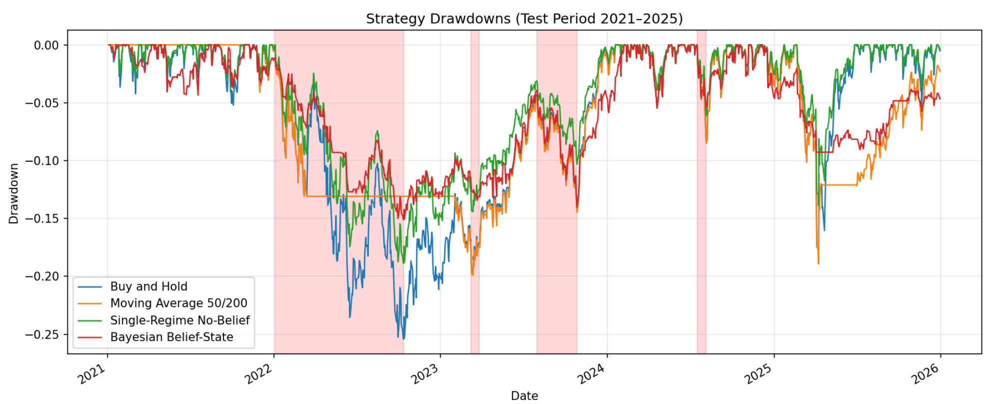

Figure 4: Drawdown comparison for the base strategies.

The base Bayesian agent has the lowest annualized volatility and lowest maximum drawdown among the main strategies. This supports the idea that belief-state modeling can make the strategy more risk-aware. However, it has the lowest total return and lowest Calmar ratio because it is conservative and trades frequently. The result is consistent with the test-period context: 2021–2025 is largely upward-trending, so being underinvested is costly. The turnover diagnostics in Table 12 quantify the trading-cost side of this issue.

The Single-Regime No-Belief strategy has the same Sharpe and Sortino ratios as Buy and Hold because it is essentially a constant scaling of the same return stream. The smaller fixed exposure lowers drawdown but does not change the mean-to-volatility ratio. This is a useful reminder that Sharpe ratio alone does not fully describe position-sizing behavior.

5.4 Belief Probabilities and Calibration

Figure 5 shows the daily belief probabilities over the three hidden regimes. The beliefs are often concentrated in one state, but they shift during volatile periods.

The belief calibration check asks whether predictive intervals contain realized returns at approximately the expected rates. Table 5 shows that the intervals are slightly conservative but close to the intended coverage levels, especially at the 90% and 95% levels.

The 91.54% empirical coverage at the 90% target is one of the strongest validation results in the project. It means that when the agent's belief-weighted predictive distribution says a realized return should fall inside

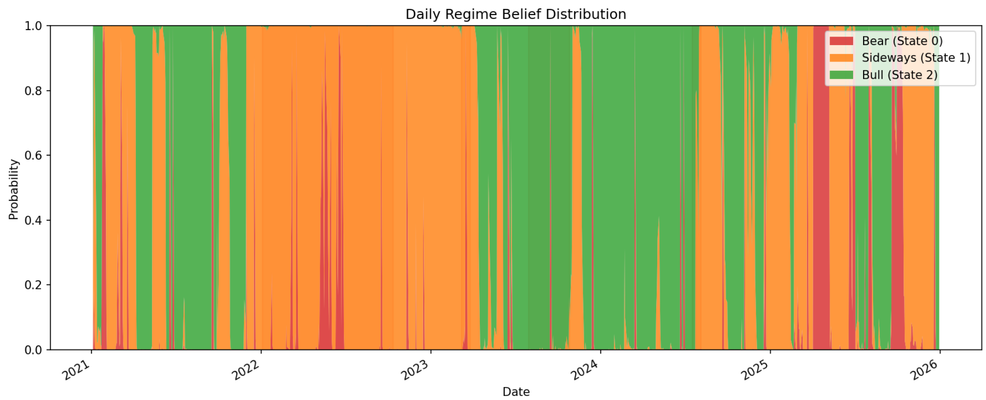

Figure 5: Daily posterior regime beliefs produced by Bayesian filtering.

Target Coverage	**Actual Coverage**	**Calibration Error**
50%	57.46%	+7.46%
75%	78.85%	+3.85%
90%	91.54%	+1.54%
95%	96.09%	+1.09%

Table 5: Belief-weighted predictive interval calibration.

a 90% interval, the realized return does so at almost the nominal rate. This makes the beliefs more than a black-box signal: they produce uncertainty estimates that are reasonably trustworthy for downstream risk decisions. The slight over-coverage at lower confidence levels reflects conservative intervals when regime beliefs are mixed, which is a safer error than overconfidence in a financial setting.

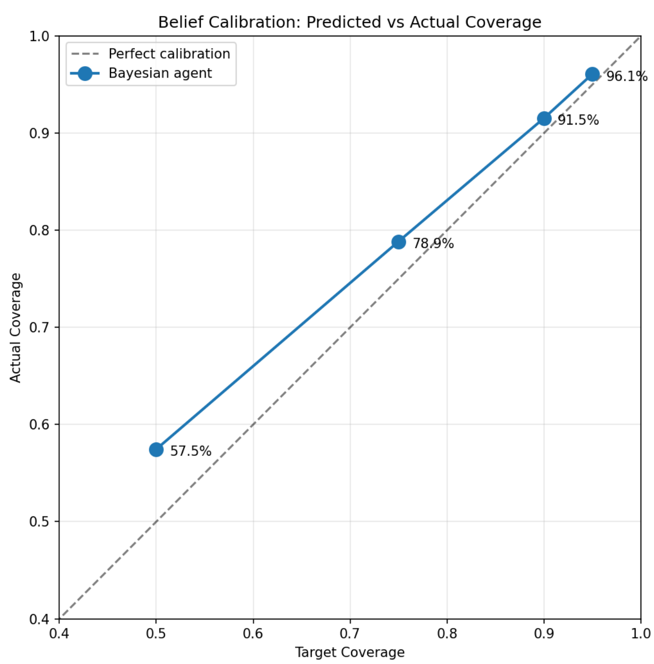

Figure 6: Calibration plot for predictive return intervals.

5.5 Performance During Stress Periods

The base Bayesian agent is most useful during some difficult periods, but the result is not uniform across every event. Table 6 summarizes normal periods and the labeled stress windows used in the evaluation. During normal periods the Bayesian agent gives up substantial upside because it is only partially invested, but it also keeps drawdown lower. During the 2022 inflation and rate-hike bear market, Buy and Hold loses 24.95%, while the Bayesian agent loses 13.94%. During the 2024 August volatility event, the Bayesian agent also loses less than Buy and Hold. The 2023 stress windows are more mixed, which is a useful reminder that belief tracking is a risk-control tool, not a guarantee of outperforming every short market event.

Period	Days	Buy & Hold Ret.	Single-Regime Ret.	Bayesian Ret.	Bayesian Max DD
Normal periods	964	198.30%	118.08%	68.36%	-9.31%
2022 Bear Market	196	-24.95%	-18.52%	-13.94%	-14.16%
2023 Banking Crisis	14	-0.22%	-0.16%	-1.79%	-2.24%
2023 Q3 Correction	64	-10.14%	-7.35%	-10.39%	-10.52%
2024 August Volatility	15	-7.90%	-5.70%	-4.76%	-5.37%

Table 6: Period-level performance by labeled market condition. The base Bayesian strategy reduces losses in the 2022 bear market and the August 2024 volatility event, but results are mixed in shorter 2023 stress windows.

The mechanism is also important. The Bayesian agent does not need to perfectly predict a crash in advance. When observations stop looking confidently bullish, posterior probability shifts toward the sideways or bear-like regimes. At the same time, the predictive variance increases through the law of total variance because the belief state is spread across regimes with different return profiles. The mean-variance optimizer then cuts exposure automatically. This explains why the strategy can reduce losses in some transition periods even when the hidden state labels are imperfect.

5.6 Entropy and Ablation Findings

Entropy is theoretically attractive because it measures uncertainty in the belief state. In practice, the entropy ablation shows that directly penalizing entropy reduces return without materially improving maximum drawdown. Table 7 summarizes the result.

Method	Total Return	Sharpe	Max DD	Avg Weight
No entropy scaling	21.57%	0.432	-15.11%	61.10%
Penalty 0.5	15.28%	0.327	-15.06%	58.30%
Penalty 1.0	10.57%	0.237	-15.00%	56.19%
Penalty 2.0	3.98%	0.095	-14.87%	53.27%
Penalty 3.0	0.27%	0.006	-15.37%	51.32%

Table 7: Entropy scaling ablation. Direct entropy penalties reduce exposure and return without meaningfully improving drawdown.

The entropy diagnostics are still useful even though direct entropy scaling is not the best trading rule. Table 8 shows that entropy measures regime ambiguity, not simply whether the market is rising or falling. Slow, persistent declines such as the 2022 bear market can have lower entropy because the observations consistently match one learned state. Sudden events, such as the August 2024 volatility shock, have higher entropy because the observations are more ambiguous across regimes.

Period	Days	Mean Entropy	Mean Normalized Entropy
Normal periods	965	0.116	0.106
2022 Bear Market	196	0.092	0.084
2023 Banking Crisis	14	0.108	0.098
2023 Q3 Correction	64	0.039	0.036
2024 August Volatility	15	0.208	0.190

Table 8: Belief entropy by market period. Entropy is highest during sudden regime ambiguity rather than during every negative-return period.

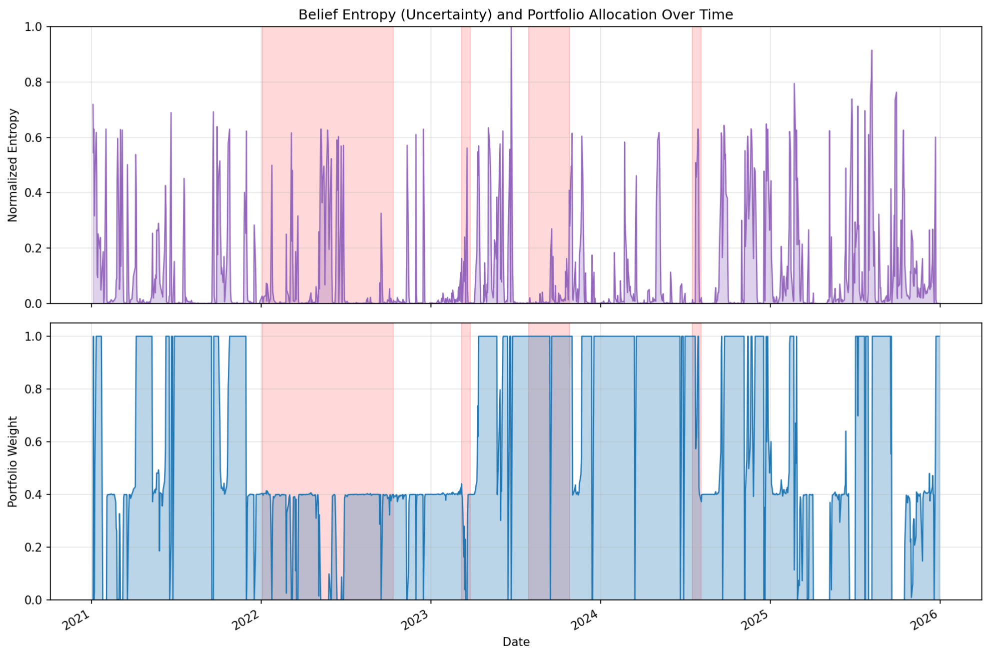

Figure 7: Belief entropy and portfolio weights over time.

Risk-aversion tuning provides a clearer improvement than direct entropy scaling. Table 9 shows the base Bayesian strategy under different λ values. Lower λ values increase market exposure but also increase drawdown. Higher λ values reduce drawdown but eventually become too conservative. In this experiment, $\lambda = 3.0$ gives the strongest Sharpe and Sortino ratios for the base mean-variance rule, while $\lambda = 2.0$ remains the main reported base setting from the original pipeline.

λ	Total Return	Sharpe	Sortino	Max DD	Avg Weight
0.5	22.02%	0.271	0.343	-33.03%	87.60%
1.0	22.90%	0.331	0.433	-26.31%	78.67%
1.5	22.48%	0.403	0.533	-18.79%	67.16%
2.0	21.57%	0.432	0.561	-15.11%	61.10%
3.0	20.83%	0.460	0.575	-11.93%	54.76%
5.0	16.56%	0.403	0.473	-10.52%	48.88%
10.0	8.18%	0.224	0.247	-10.09%	42.35%

Table 9: Risk-aversion sweep for the base Bayesian mean-variance strategy.

Feature ablations show that Volume is not helpful in the current setup. The Log_Return + Volatility feature set produces stronger risk-adjusted performance than the full three-feature set for the base mean-variance strategy. This finding motivates the enhanced strategy's feature choice.

Features	Total Return	Sharpe	Max DD	Avg Weight
Log_Return only	26.12%	0.447	-12.51%	70.38%
Log_Return + Volume	-5.34%	-0.154	-17.75%	33.61%
Log_Return + Volatility	22.15%	0.552	-8.54%	44.98%
All three features	21.57%	0.432	-15.11%	61.10%

Table 10: Feature ablation for the Bayesian strategy.

5.7 Enhanced Bear-Defense Strategy

The strongest practical improvement is the enhanced bear-defense rule. The full-period enhanced sweep gives a descriptive result of 70.59% total return, Sharpe 0.771, Sortino 0.987, and maximum drawdown -20.56% over 2021–2025. However, because that sweep uses the full period for exploration, the more defensible result is the validation-selected held-out check shown in Table 11.

Strategy	Total Return	Ann. Return	Ann. Vol.	Sharpe	Sortino	Max DD	Avg Weight	Cost
Buy and Hold	80.33%	19.65%	15.03%	1.308	1.726	-18.90%	100.00%	0.00%
Moving Average 50/200	43.48%	12.11%	14.10%	0.859	1.057	-18.90%	90.15%	0.30%
Single-Regime No-Belief	52.30%	14.02%	10.72%	1.308	1.726	-13.88%	71.34%	0.00%
Enhanced Bear-Defense Bayesian	60.78%	15.80%	12.88%	1.227	1.573	-15.85%	94.72%	0.20%

Table 11: Held-out 2023–2025 results after selecting the enhanced configuration on 2021–2022 validation data.

A light statistical check supports this cautious interpretation. A nonparametric bootstrap with 5,000 resamples of held-out daily returns gives a 95% Sharpe interval of [0.14, 2.41] for Buy and Hold and [0.11, 2.41] for the enhanced strategy. The paired bootstrap interval for the enhanced-minus-Buy-and-Hold Sharpe difference is [-0.46, 0.33], so the held-out Sharpe difference is not statistically decisive.

The enhanced strategy also addresses the transaction-cost problem observed in the original Bayesian agent. Table 12 reports the full-period exploratory turnover diagnostics. The original Bayesian agent changes exposure frequently, producing high turnover and a large cost drag. The enhanced bear-defense rule trades much less often because it stays near full exposure unless the bear-state belief becomes elevated.

Strategy	Total Turnover	Trades	Cost Drag	Avg Weight
Buy and Hold	0.00	0	0.00%	100.00%
Single-Regime No-Belief	0.00	0	0.00%	71.34%
Original Bayesian	81.73	656	8.17%	61.10%
Enhanced Bayesian	8.34	60	0.83%	89.41%

Table 12: Full-period exploratory turnover and transaction-cost diagnostics. The enhanced strategy sharply reduces daily rebalancing compared with the original Bayesian agent.

In the held-out 2023–2025 window, the enhanced strategy does not beat Buy and Hold on total return or risk-adjusted ratios. It does, however, earn higher total return than the Moving Average and Single-Regime baselines, while keeping lower volatility and drawdown than Buy and Hold. Compared with the original Bayesian design, the practical improvement is that it stays invested most of the time and avoids daily rebalancing. The result supports the project goal in a careful way: beliefs are most useful as a risk-control signal, not as a guarantee of maximum raw return.

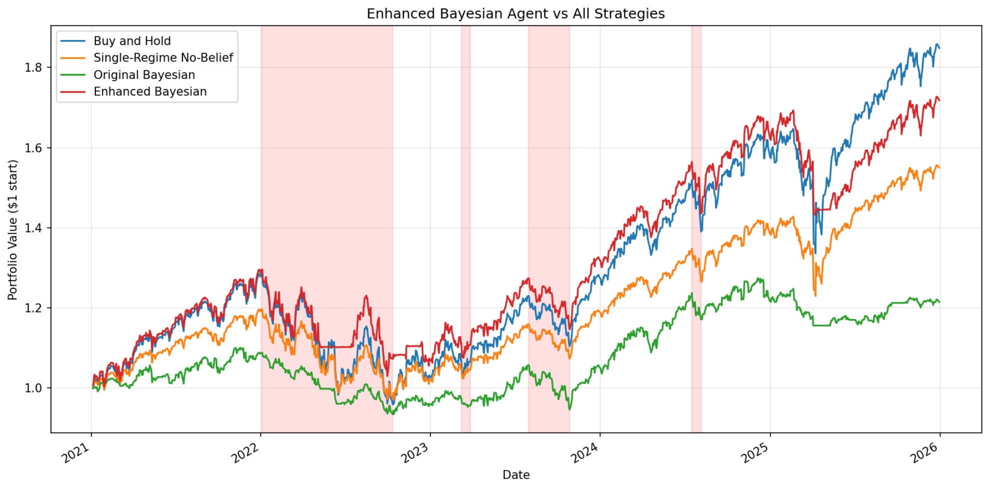

Figure 8: Full-period enhanced strategy equity-curve comparison. The full-period sweep was used for exploration only; Table 11 reports the unbiased held-out 2023–2025 evaluation.

5.8 `pgmpy` Dynamic Bayesian Network Check

The pgmpy component represents the regime process as a Dynamic Bayesian Network with a temporal regime edge and observed feature nodes. Because the continuous Gaussian likelihoods are learned by the HMM, the DBN result is best interpreted as a representation check rather than a replacement model.

The DBN has two time slices. Within a slice, the hidden node Regime_t points to the observed feature nodes Log_Return_t, Volume_t, and Volatility_t. Across slices, the temporal edge $\text{Regime}_t \rightarrow \text{Regime}_{t+1}$ encodes the Markov transition. Forward inference in this DBN with the same Gaussian conditional distributions is mathematically equivalent to the HMM forward filter, so the pgmpy version mainly makes the graphical-model structure explicit.

Using the same transition and Gaussian emission parameters, the DBN-style forward update agrees with the HMM belief filter on 99.92% of most-likely regime labels. The mean absolute belief-probability difference is 0.0006. This high agreement is expected because the two methods implement the same Bayesian filtering idea; the small remaining difference mainly comes from initialization, since the main HMM filter starts from a uniform prior while the DBN check starts from the fitted HMM start probabilities. The DBN serves as a structural representation check, not as independent empirical evidence.

6 Discussion

6.1 Main Findings

The project produces three main findings.

First, the base Bayesian belief-state agent is risk-aware. It lowers volatility and maximum drawdown relative to the baselines, but it is too conservative and suffers from high turnover costs. The calibration results also support the belief-state interpretation: the agent's predictive intervals are close to their intended coverage levels, so the posterior beliefs are not just arbitrary trading scores. This shows that belief tracking helps with risk control but does not automatically produce high returns.

Second, entropy is useful as a diagnostic, but direct entropy-based exposure scaling is not the strongest improvement in this test period. The law-of-total-variance portfolio rule already makes the agent more cautious when regime beliefs are uncertain.

Third, the enhanced bear-defense strategy is a better practical use of the belief state. It stays invested by default and reduces exposure only when posterior bear probability rises. This preserves much more upside than the base Bayesian agent while still using Bayesian beliefs for risk control.

6.2 Alignment with the Project Plan

The final implementation is aligned with the proposal. It includes hidden regimes, HMM parameter estimation, Bayesian belief updates, belief-conditioned portfolio decisions, transaction costs, baselines, and out-of-sample evaluation. The `pgmpy` work also addresses the planned Bayesian-network extension. The main difference from the initial plan is that the final project is more careful about validation: the enhanced strategy is reported with a 2021–2022 validation split and a 2023–2025 held-out check.

6.3 Limitations

Several limitations remain:

- **Single asset:** the agent trades only the S&P 500 against cash. A multi-asset version could allocate across equities, bonds, commodities, or sectors.
- **Fixed HMM parameters:** the HMM is trained once on 2015–2020 and is not updated online.
- **Memoryless regimes:** the HMM transition model does not depend on how long the market has been in a regime.
- **K selection:** higher K values improve AIC/BIC, so $K = 3$ should be viewed as interpretable rather than statistically best. The marginal AIC/BIC improvement does not flatten in the swept range, so a fully data-driven choice would prefer $K \geq 8$.
- **COVID training effect:** the training-period COVID crash makes the bear-like state very extreme, so milder bear markets may not trigger that state strongly.
- **Enhanced feature set:** the enhanced strategy uses Log_Return + Volatility, not the full three-feature HMM. This is justified by ablation results, but it means the enhanced result combines a strategy change with a feature-selection change.
- **Limited robustness evidence:** the held-out 2023–2025 evaluation provides a useful check against tuning directly on the full test period, but it is still only one market history. A stronger robustness study would use rolling train-validation-test windows across multiple historical periods and report uncertainty bands or confidence intervals for the performance metrics.

6.4 Ethical Considerations

Automated trading systems can influence market behavior if deployed at scale. If many algorithms respond to similar regime signals, they may amplify selling during stress periods. Backtests can also create false confidence, especially when a strategy is tuned on a limited historical period. There is also a transparency concern because the HMM states are latent and only receive bear, sideways, and bull labels after training. A separate fairness concern is that sophisticated quantitative tools are easier for institutions to access than ordinary retail investors. However, this implementation uses publicly available daily data and standard Python libraries, making it reproducible with modest computational resources. For these reasons, the results should be interpreted as a research prototype, not as investment advice. Any real deployment would require additional validation, risk controls, and transparency.

6.5 Future Work

Natural extensions include:

- extending the agent from a single index-and-cash setting to a multi-asset portfolio with stocks, bonds, sectors, or volatility hedges;
- refitting or updating HMM parameters over time so the regime model can adapt to new market conditions;
- replacing the memoryless HMM with a duration-aware or semi-Markov model;
- adding macro or market-stress variables such as interest rates, inflation, VIX, or sector returns as extra DBN nodes;
- running rolling train-validation-test experiments and reporting uncertainty bands for Sharpe, Sortino, and drawdown.

7 Conclusion

This project implements a complete Bayesian belief-state trading pipeline. The final system estimates hidden market regimes with an HMM, updates posterior regime beliefs through Bayesian filtering, and uses those beliefs to make risk-aware portfolio decisions. The base Bayesian agent reduces volatility and drawdown but gives up return because it is conservative and expensive to rebalance. The enhanced bear-defense strategy shows a more practical use of the belief state: remain invested most of the time, but reduce exposure when posterior bear-regime probability becomes elevated.

The final conclusion is mixed but useful. In a bull-heavy period, Buy and Hold remains difficult to beat on total return. However, belief-state modeling provides a principled way to represent uncertainty, evaluate risk, and reduce downside exposure. The project therefore supports the value of Bayesian regime beliefs for risk-aware autonomous trading, while also showing that portfolio design and transaction-cost control are just as important as the regime model itself.

Reproducibility Statement

All tables and figures are generated from the local pipeline in `belief_state_trader/`. The project repository is available at rowe0137/5512-Final-Stock-Trader. From `belief_state_trader/`, the main command is:

```
python scripts/run_all.py.
```

Report source files are in `reports/`, generated CSVs and plots are in `results/`, and dependencies are listed in `requirements.txt`. The raw Yahoo Finance pull is documented in `scripts/00_data_collection.ipynb`. The only stochastic step is HMM initialization: the model is fit with 10 random restarts using seeds 42–51, and the best training log-likelihood is kept. Given that fitted HMM and the committed data, filtering, portfolio decisions, baselines, ablations, and held-out evaluation are deterministic. This makes the reported numbers reproducible without hidden manual steps.

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